Radome – General data

1. General

The radome is installed on forward surface of the nose baggage compartment structure.

2. Description

The DHC-6 Series 400 retains the original DeHavilland DHC-6-300 radome design with the exception the Series 400 radome contains four lightning diverter strips as opposed to two.

The radome is attached to the nose using NAS 8403 (10 – 32) screws, four of which participate in connecting the lightning diverters to the bonding straps.

The nose assembly includes a continuation of the lightning diverting function of the radome diverters. The radome lightning diverters are attached the nose assembly's bonding straps and carbon fibre structure in a lap joint using NAS 8403 (10 – 32) screws. The bonding straps are also attached to the nose assembly using two blind rivets. The bonding straps channel the lightning current into the copper foil of the nose whereupon it is spread through the nose structure and into the aluminum fuselage.

At each of the bonding strap locations, an added local "reinforcement" ply of expanded copper foil is applied to provide additional foil cross section to better manage the current density that could occur in diverting the energy from a lightning strike. The bonding strap in both instances will provide a significant "footprint" for the conduction of lightning currents into the underlying expanded foil and carbon fibre structure.

3. Adjustment/Test

A. Radome Bonding Test

Measure the resistance between the radome diverter strips and the airframe using a milliohm meter, resistance to be less than 10 milliohms. Probe measurements to be taken on the radome diverter strip and a corresponding airframe side bonding strap attachment rivet. Repeat for each diverter strip (quantity 4 places).

B. Radome Bonding Adjustment/Rework

Remove the radome diverter strip to bonding strap attachment screw and clean as required to achieve bonding requirement. If required, remove the radome and clean the mating surface of the nose assembly bonding strap in accordance with standard procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

4. Removal and Installation

A. Removal of Radome

1) Remove radome fasteners and remove the radome from the forward nose assembly.

B. Installation of Radome

1) Set radome in position and install fasteners.

2) Ensure disconnected diverter strip to bonding strap screws are installed.

3) Verify electrical bonding of the radome to airframe is maintained. Refer to Para 3.A.

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